

PeakFocus: An End-to-End Framework Bridging Localization and Regression for Electricity Load Peak Forecasting

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Abstract

Electricity load peak forecasting (ELPF), simultaneously predicting peak timing and intensity, is a prerequisite for effective grid scheduling and risk management. However, existing methods adopt a two-stage predict-then-locate paradigm, which severs the link between temporal localization and numerical regression. Second, insufficient representation of multi-scale peak patterns leads to peak misjudgment and timing misalignment. Third, the lack of explicit perception during peak intensity forecasting causes peak intensity to be diluted by global smoothing trends. To address these limitations, we propose PeakFocus, an end-to-end framework for ELPF. (i) A Unified Peak-Aware Pipeline (UPAP) utilizes a triple hybrid loss to synergize timing and intensity forecasting, alongside a tolerance-based evaluation protocol. (ii) A Multi-Scale Mixing Peak Locator (MSM-PL) exploits coarse-grained features to mitigate misjudgments caused by local fluctuations, and injects them into fine-grained features via a cascade mechanism to resolve misalignment. (iii) A Location-Aware Decoder (LAD) injects peak timing context into the intensity prediction process, providing explicit guidance to counteract smoothing and optimize peak intensity. Extensive evaluations of both an open-source load dataset and a world’s largest electricity load dataset substantiate that PeakFocus outperforms baselines in both timing precision and intensity estimation. Code is available at <https://anonymous.4open.science/r/PeakFocus>.

1. Introduction

Time series forecasting (TSF) is pivotal across various domains, ranging from energy management, finance, and economics (Benidis et al., 2022; Jiang et al., 2025). While deep learning architectures like Transformers (Zhou et al., 2021; Wu et al., 2021) and Multi-Layer Perceptrons (MLPs) (Wang et al., 2024a; Hu et al., 2025a) effectively model temporal dynamics, they typically rely on global reconstruction

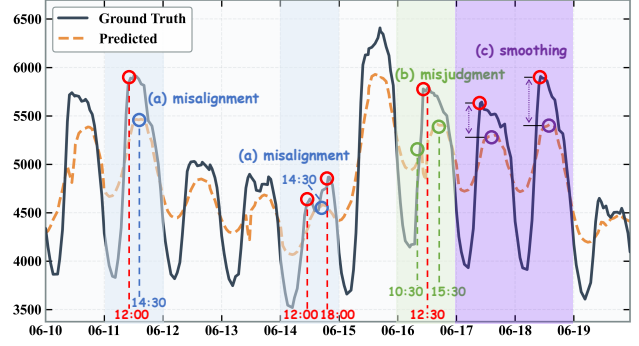


Figure 1. An illustration of limitations in PatchTST for ELPF. (a) Timing Misalignment and (b) Peak Misjudgment illustrate the multi-scale representation conflict, where coarse-grained features lack temporal resolution (resulting in shifts) while fine-grained features are susceptible to local fluctuations (leading to false positives). (c) Intensity Smoothing shows the underestimation caused by the lack of explicit peak timing context, where the intensity decoder operates blindly and is diluted by global smoothing trends.

losses such as Mean Squared Error (MSE) or Mean Absolute Error (MAE) (Wang et al., 2024b). This optimization biases the model toward dominant trends while neglecting sparse peak signals, failing to meet the practical demands of extreme event forecasting. For instance, effective power grid management requires precise prediction of peak timing and intensity rather than smoothed trajectories. Thus, Electricity Load Peak Forecasting (ELPF) has emerged as a highly specialized task designed to address these critical needs (Amara-Ouali et al., 2023; Huang et al., 2025).

Existing ELPF methods face distinct structural limitations inherent to their paradigms (Dai et al., 2021). First, when applying TSF models to capture peak features, they typically rely on a two-stage predict-then-locate paradigm. This separation severs the link between temporal localization and intensity regression, preventing the model from learning their joint relationship effectively. To bridge this gap, recent end-to-end methods have emerged (Zhang et al., 2023; Zhu et al., 2025b). However, they still struggle to resolve the multi-scale representation conflict. Coarse-grained features lack the necessary temporal resolution, resulting in timing misalignment (Figure 1a), while fine-grained features are susceptible to local fluctuations, leading to peak misjudgment (Figure 1b). On the other hand, the intensity prediction suffers from a lack of explicit timing context.

Even in current unified frameworks, the intensity decoder operates blindly without direct guidance on when a peak occurs. Consequently, predictions are inevitably diluted by global smoothing trends (Figure 1c), resulting in a systematic underestimation of peak intensity.

To address these limitations, we propose PeakFocus, an end-to-end framework specifically engineered to simultaneously unify peak localization and intensity regression. (i) To bridge the paradigm disconnect, we introduce the Unified Peak-Aware Pipeline (UPAP). UPAP utilizes a triple hybrid loss function to synergistically optimize both timing and intensity, ensuring the model effectively learns their joint distribution. (ii) To resolve the multi-scale conflict, the Multi-Scale Mixing Peak Locator (MSM-PL) exploits coarse-grained semantics to suppress local fluctuations and prevent misjudgment, while simultaneously injecting this robust context into fine-grained features via a top-down cascade mechanism to achieve precise temporal alignment. (iii) To counteract intensity smoothing, the Location-Aware Decoder (LAD) conditions the forecasting process on peak timing context, providing the explicit guidance required to accurately estimate true peak intensity without dilution, thereby improving overall prediction accuracy. Our main contributions are summarized as follows:

- **End-to-End Peak Forecasting.** We articulate a fundamental paradigm shift from conventional global reconstruction to sparse extreme value optimization. By synergizing temporal localization and numerical regression, our framework unifies peak timing localization and intensity regression. This methodology enables the direct learning of critical peak signals typically suppressed by standard global objectives.
- **Peak-Aware Methodology.** We design specialized components to tackle intrinsic structural conflicts. Specifically, the MSM-PL resolves the trade-off between misjudgment and misalignment via multi-scale mixing, while the LAD employs context injection to overcome the intensity smoothing dilemma.
- **Comprehensive Evaluation Protocol.** We establish a tolerance-aware evaluation protocol to assess both localization accuracy and intensity regression error. Furthermore, we conduct extensive evaluations on an open-source electricity dataset (ELC) and the world largest electricity load dataset (WLEL). Results substantiate that PeakFocus outperforms baselines in both timing precision and intensity estimation.

2. Preliminaries and Problem Definition

2.1. Peak Definition

We characterize peak events based on local optimality and intensity prominence. Given the ground-truth load intensity series $\mathbf{Y}_{i,\text{true}} \in \mathbb{R}^{H \times 1}$, we employ the standard peak

detection algorithm from the open-source Python library `findpeaks`¹. Formally, a time step t is labeled as a peak ($\mathbf{Y}_{t,\text{true}}[t] = 1$) if it represents a valid local maximum satisfying specific prominence constraints defined by a sensitivity threshold η and a lookahead window ℓ . The detailed generation process is provided in Appendix A.

2.2. ELPF Task Formulation

The goal of ELPF is to jointly anticipate the timing and intensity of extreme events. Formally, given a historical univariate load series $\mathbf{X}_{1:T} \in \mathbb{R}^{T \times 1}$, the model learns a mapping function $\mathcal{F}_\theta$ that produces two aligned outputs over a horizon H , where $\mathbf{Y}_{i,\text{pred}} \in \mathbb{R}^{H \times 1}$ denotes the predicted intensity trajectory, and $\mathbf{Y}_{t,\text{pred}} \in [0, 1]^{H \times 1}$ represents the predicted peak timing probabilities:

$$[\mathbf{Y}_{i,\text{pred}}, \mathbf{Y}_{t,\text{pred}}] = \mathcal{F}_\theta(\mathbf{X}_{1:T}, \mathbf{M}_{1:T+H}) \quad (1)$$

3. Methodology

As illustrated in Figure 2, PeakFocus is designed as a dual-head architecture underpinned by a shared backbone. The process begins with the Encoder (Section 3.1), which applies RevIN normalization and embedding to raw inputs $\mathbf{X}_{1:T}$ and temporal features $\mathbf{M}_{1:T}$ to construct a horizon-aligned context. To simultaneously capture both timing and intensity, the architecture branches into two specialized streams. First, the MSM-PL (Section 3.2) injects coarse-grained semantics into fine-grained features, producing the peak hidden state $\mathbf{H}_{\text{pl}}$ and timing probabilities $\mathbf{Y}_{t,\text{pred}}$. Leveraging this context, the LAD (Section 3.3) explicitly conditions the intensity forecasting $\mathbf{Y}_{i,\text{pred}}$ on $\mathbf{H}_{\text{pl}}$ via location-aware cross-attention to prevent intensity dilution. Finally, the entire framework is optimized end-to-end via the UPAP (Section 3.4) using a triple hybrid objective under consistent tolerance-aware semantics for enhanced precision.

3.1. Normalization and Encoder

As illustrated in Figure 2, we first apply RevIN normalization (Kim et al., 2021) to the raw input $\mathbf{X}_{1:T} \in \mathbb{R}^{T \times 1}$ to obtain the normalized series:

$$\mathbf{X}_{\text{norm},1:T} = \text{RevIN}(\mathbf{X}_{1:T}) \quad (2)$$

The input data for the encoder $\mathbf{X}_{\text{en},\text{in}}$ are generated by the Data Embedding Layer (Zhou et al., 2021), including value projection, positional encoding, and temporal features:

$$\mathbf{X}_{\text{en},\text{in}} = \mathcal{E}_{\text{val}}(\mathbf{X}_{\text{norm},1:T}) + \mathcal{E}_{\text{pos}} + \mathcal{E}_{\text{time}}(\mathbf{M}_{1:T}) \quad (3)$$

where $\mathcal{E}_{\text{val}}(\cdot)$ denotes the 1D convolutional value embedding, $\mathcal{E}_{\text{pos}}$ represents the fixed positional encoding, and $\mathcal{E}_{\text{time}}(\cdot)$ captures timestamp features $\mathbf{M}$.

¹<https://github.com/erdogant/findpeaks>

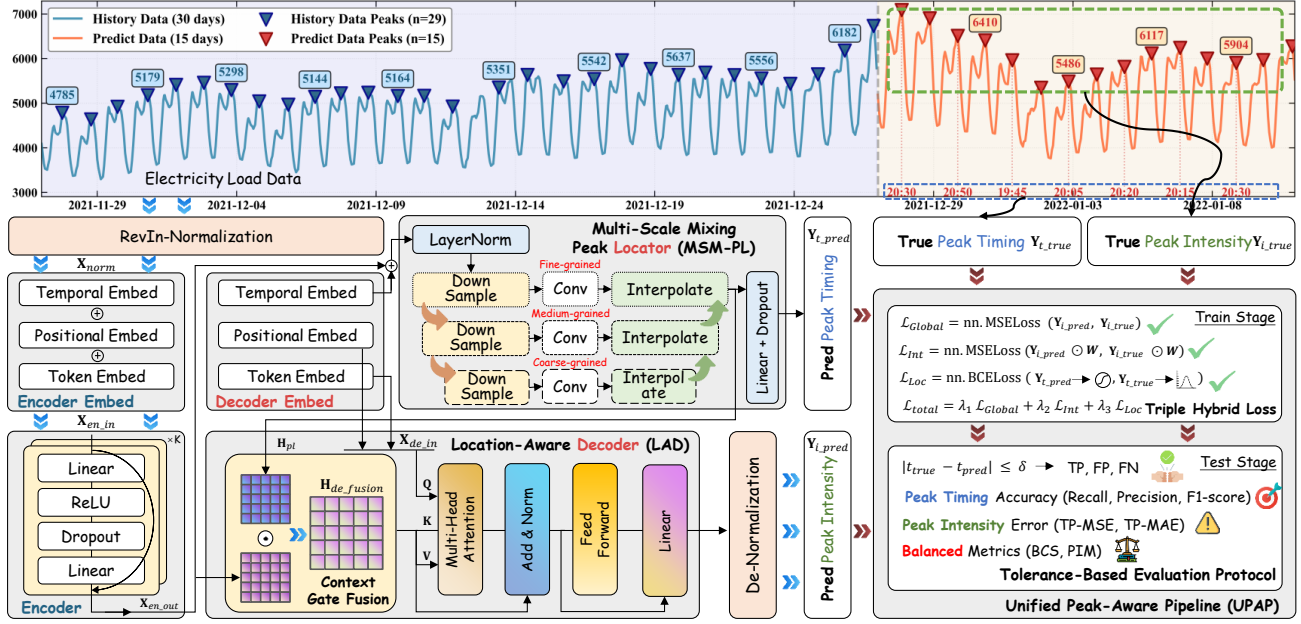


Figure 2. The PeakFocus architecture. An encoder extracts input features. MSM-PL resolves localization conflicts via multi-scale mixing, outputting peak states $\mathbf{H}_{pl}$ and timing $\mathbf{Y}_{t,pred}$. LAD prevents intensity smoothing by conditioning regression $\mathbf{Y}_{i,pred}$ on $\mathbf{H}_{pl}$ via Context Gate Fusion. UPAP ensures robust optimization via a Triple Hybrid Objective ($\mathcal{L}_{total}$) under tolerance-aware semantics.

To extract deep temporal patterns and align features with the forecasting horizon, we further employ a residual MLP backbone f_{enc} to obtain robust latent representations $\mathbf{Z}_{hist}$ (Shao et al., 2022), followed by a learnable temporal projection Π . This process yields horizon-aligned context $\mathbf{X}_{en,out}$.

$$\mathbf{Z}_{hist} = f_{enc}(\mathbf{X}_{en,in}) \in \mathbb{R}^{T \times d} \quad (4)$$

$$\mathbf{X}_{en,out} = \Pi(\mathbf{Z}_{hist}) \in \mathbb{R}^{H \times d} \quad (5)$$

3.2. Multi-Scale Mixing Peak Locator

MSM-PL is designed to resolve the intrinsic trade-off between peak misjudgment and timing misalignment. We incorporate future temporal information into the peak branch by injecting time features into the shared context (Hu et al., 2025a; Zhu et al., 2025c), followed by layer normalization (Wu et al., 2024) to stabilize the feature distribution:

$$\mathbf{H}_{in}^p = \text{LayerNorm}(\mathbf{X}_{en,out} + \mathcal{E}_{time}(\mathbf{M}_{T+1:T+H})) \quad (6)$$

To mitigate misjudgments caused by local fluctuations, we employ a hierarchical architecture comprising a bottom-up pyramid and a top-down cascade. Inspired by (Wu et al., 2021), we initially construct a hierarchy of coarse-grained features through iterative downsampling. Let $\text{AvgPool}_{\kappa,\sigma}(\cdot)$ denote the average pooling operation parameterized by kernel size κ and stride σ . The pyramid generation is formally defined as:

$$\mathbf{S}^{(0)} = \mathbf{H}_{in}^p \quad \mathbf{S}^{(k)} = \text{AvgPool}_{\kappa^{(k)},\sigma^{(k)}}(\mathbf{S}^{(k-1)}) \quad (7)$$

To leverage these robust semantics for precise alignment, we fuse the coarse representations back into finer scales. This is achieved via a cascade injection mechanism based on a convolutional layer $\psi_k(\cdot)$ (with kernel size κ_c) and an average interpolation upsampling operator $\mathcal{U}_k(\cdot)$ (Chow & Lin, 1971). The cascade generation is performed recursively for $k = K - 1, \dots, 0$ as:

$$\mathbf{F}^{(K)} = \psi_K(\mathbf{S}^{(K)}) \quad (8)$$

$$\mathbf{F}^{(k)} = \psi_k(\mathbf{S}^{(k)}) + \mathcal{U}_k(\mathbf{F}^{(k+1)}) \quad (9)$$

The resulting finest fused feature $\mathbf{F}^{(0)}$ constitutes the peak hidden state $\mathbf{H}_{pl} \in \mathbb{R}^{H \times d}$, serving as critical context for the downstream decoder. We obtain the peak timing probability $\mathbf{Y}_{t,pred}$ by projecting this representation through a linear classifier parameterized by weight $\mathbf{W}_p \in \mathbb{R}^{d \times 1}$ and bias $\mathbf{b}_p \in \mathbb{R}$, followed by the sigmoid activation function $\sigma(\cdot)$:

$$\mathbf{Y}_{t,pred} = \sigma(\mathbf{H}_{pl}\mathbf{W}_p + \mathbf{b}_p) \in [0, 1]^{H \times 1} \quad (10)$$

3.3. Location-Aware Decoder

To counteract the intensity smoothing dilemma, the LAD explicitly conditions the regression process on the peak hidden state $\mathbf{H}_{pl}$ via a location-aware cross-attention mechanism. We first construct decoder queries $\mathbf{X}_{de,in}$ by embedding normalized values with positional encoding:

$$\mathbf{X}_{de,in} = \mathcal{E}_{val}(\mathbf{X}_{dec}) + \mathcal{E}_{pos} \in \mathbb{R}^{H \times d} \quad (11)$$

where $\mathbf{X}_{dec}$ represents the decoder input values and $\mathcal{E}_{pos}$ provides the localized context. Simultaneously, to inject

peak timing information into the forecasting context, we employ a Context Gate Fusion mechanism to modulate the shared horizon representation $\mathbf{X}_{\text{en.out}}$ using the peak hidden state $\mathbf{H}_{\text{pl}}$, where $\odot$ denotes element-wise multiplication:

$$\mathbf{G} = \tanh(\mathbf{H}_{\text{pl}}) \odot \sigma(\mathbf{X}_{\text{en.out}}) \quad (12)$$

Subsequently, a multi-head cross-attention mechanism is employed to guide intensity decoding (Vaswani et al., 2017), where $\mathbf{X}_{\text{de.in}}$ serves as the Query and the gated context $\mathbf{G}$ serves as both Key and Value. For each head $h \in \{1, \dots, H_a\}$, the calculation is performed as:

$$\mathbf{Q}^{(h)} = \mathbf{X}_{\text{de.in}} \mathbf{W}_Q^{(h)} \quad (13)$$

$$\mathbf{K}^{(h)} = \mathbf{G} \mathbf{W}_K^{(h)} \quad (14)$$

$$\mathbf{V}^{(h)} = \mathbf{G} \mathbf{W}_V^{(h)} \quad (15)$$

where $\mathbf{W}_Q^{(h)}, \mathbf{W}_K^{(h)}, \mathbf{W}_V^{(h)} \in \mathbb{R}^{d \times d_h}$. Then, the aggregated attention output $\mathbf{O}_{\text{attn}}$ is obtained by concatenating all head outputs and projecting them afterward:

$$\mathbf{O}^{(h)} = \text{softmax}\left(\frac{\mathbf{Q}^{(h)}(\mathbf{K}^{(h)})^\top}{\sqrt{d_h}}\right) \mathbf{V}^{(h)} \quad (16)$$

$$\mathbf{O}_{\text{attn}} = \text{Concat}(\mathbf{O}^{(1)}, \dots, \mathbf{O}^{(H_a)}) \mathbf{W}_O \quad (17)$$

Following the decoder layer, the attention output is processed through a residual connection, normalization, and a Feed-Forward Network (Vaswani et al., 2017):

$$\mathbf{Z} = \text{LayerNorm}(\mathbf{X}_{\text{de.in}} + \mathbf{O}_{\text{attn}}) \quad (18)$$

$$\mathbf{H}_{\text{dec}} = \text{LayerNorm}(\mathbf{Z} + \text{FFN}(\mathbf{Z})) \quad (19)$$

where $\text{FFN}(\cdot)$ consists of 1D convolutions with ReLU activation (Weng et al., 2018). Finally, the decoder is projected to generate the normalized intensity, which is then inverted via RevIN to produce the final forecast $\mathbf{Y}_{i.\text{pred}}$:

$$\tilde{\mathbf{Y}}_{i.\text{pred}} = \text{Proj}(\mathbf{H}_{\text{dec}}) \quad (20)$$

$$\mathbf{Y}_{i.\text{pred}} = \text{DeNorm}(\tilde{\mathbf{Y}}_{i.\text{pred}}) \in \mathbb{R}^{H \times 1} \quad (21)$$

3.4. Unified Peak-Aware Pipeline

Triple Hybrid Loss Function. PeakFocus is trained end-to-end using a triple hybrid objective:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{Global}} + \lambda_2 \mathcal{L}_{\text{Int}} + \lambda_3 \mathcal{L}_{\text{Loc}} \quad (22)$$

To ensure general trend consistency and guarantee basic forecasting capability, we first employ the standard Mean Squared Error (MSE) loss on the global view:

$$\mathcal{L}_{\text{Global}} = \text{MSE}(\mathbf{Y}_{i.\text{pred}}, \mathbf{Y}_{i.\text{true}}) \quad (23)$$

To mitigate peak sparsity and introduce tolerance awareness for the subsequent peak-specific objectives, we construct a soft weight mask $\mathcal{W} \in [0, 1]^{H \times 1}$ based on the ground-truth peak indices $\mathcal{P} = \{k \mid p_k = 1\}$. Specifically, the mask values are generated using a truncated Gaussian kernel (Tobar et al., 2015), as shown below:

$$\mathcal{W} = \max_{k \in \mathcal{P}} \exp\left(-\frac{(t-k)^2}{2\gamma^2}\right) \mathbb{I}(|t-k| \leq \delta) \quad (24)$$

Leveraging this mask, we define a peak-aware intensity objective to explicitly guide the model in capturing peak magnitude. By scaling the gradients with the squared mask $\mathcal{W}^2$, this objective prioritizes high-confidence peak regions and imposes a strict focus on the peak apexes, complementing the trend supervision provided by $\mathcal{L}_{\text{Global}}$:

$$\mathcal{L}_{\text{Int}} = \text{MSE}(\mathbf{Y}_{i.\text{pred}} \odot \mathcal{W}, \mathbf{Y}_{i.\text{true}} \odot \mathcal{W}) \quad (25)$$

Finally, we apply the same masking strategy to the temporal dimension for robust localization. Specifically, treating $\mathcal{W}$ as the soft label for peak probability, we employ the Binary Cross-Entropy (BCE) loss (Fulcher & Jones, 2014) to provide spatially continuous supervision and ensure effective learning, as shown below:

$$\mathcal{L}_{\text{Loc}} = \text{BCE}(\mathbf{Y}_{t.\text{pred}}, \mathcal{W}) \quad (26)$$

Tolerance-based Evaluation Protocol. Distinct from the soft-target training strategy, accurate assessment of peak forecasting requires mapping continuous probability outputs to discrete events. To achieve this, we establish a Tolerance-based Condense-and-Match Protocol to rigorously evaluate both localization and intensity performance.

Condense-and-Match Mechanism. To bridge continuous probabilities and discrete metrics, we extract predicted peak candidates $\hat{\mathcal{P}}$ by identifying local maxima in $\mathbf{Y}_{t.\text{pred}}$ exceeding a threshold τ . We then construct the set of True Positive pairs $\mathcal{M}$ via a strict one-to-one bipartite matching between $\hat{\mathcal{P}}$ and ground-truth peaks $\mathcal{P}$ within a temporal tolerance δ . Only successfully matched pairs $\mathcal{M} = \{(t, \hat{t}) \mid |t - \hat{t}| \leq \delta\}$ are used for subsequent intensity error calculation (see Appendix B for the complete algorithm).

Peak Timing Metrics. To evaluate localization precision, we report standard classification metrics (Fulcher & Jones, 2014) derived from the matching results (i.e., unmatched items denote False Positives or False Negatives):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (27)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (28)$$

$$\mathcal{F}_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (29)$$

Peak Intensity Error. We assess regression quality exclusively on the matched set $\mathcal{M}$ (i.e., the True Positive pairs). This design ensures that intensity errors are computed only between semantically aligned peak events, decoupling intensity estimation performance from localization failures:

$$\text{TP-MSE} = \frac{1}{|\mathcal{M}|} \sum_{(t,i) \in \mathcal{M}} ((\mathbf{Y}_{i\text{-pred}})_{\hat{t}} - (\mathbf{Y}_{i\text{-true}})_t)^2 \quad (30)$$

$$\text{TP-MAE} = \frac{1}{|\mathcal{M}|} \sum_{(t,i) \in \mathcal{M}} |(\mathbf{Y}_{i\text{-pred}})_{\hat{t}} - (\mathbf{Y}_{i\text{-true}})_t| \quad (31)$$

Balanced Metrics. To provide a comprehensive assessment, we report the Balanced Composite Score (BCS) and Peak-Intensity Magnification (PIM), which combine the unmasked TP-MSE with the localization $\mathcal{F}_1$:

$$\text{BCS} = \alpha(1 - \mathcal{F}_1) + (1 - \alpha) \left(1 - \frac{1}{1 + \text{TP-MSE}} \right) \quad (32)$$

$$\text{PIM} = 1 + \frac{\text{TP-MSE}}{\mathcal{F}_1 + \epsilon} \quad (33)$$

4. Experiments

4.1. Experimental Setup

Datasets. We make evaluations on two real-world datasets: (1) **Electricity (ELC)** (Zhou et al., 2021): A public benchmark containing hourly samples from 2016-07 to 2019-07. We follow a 6:2:2 split ratio: the training set covers 2016-07 to 2018-04, the validation set covers 2018-05 to 2018-11, and the test set covers 2018-12 to 2019-07. (2) **World-Largest Electricity Load (WLEL)**: A massive industrial dataset recorded at hourly intervals from 2021-01 to 2025. It is chronologically split into training (2021-01 to 2023-08), validation (2023-09 to 2024-08), and testing (2024-09 to 2025-09) sets to evaluate long-term robustness.

Baselines. We evaluate PeakFocus against two categories of SOTA models: TSF models and ELPF models. The TSF baselines include Transformer (Vaswani et al., 2017), Informer (Zhou et al., 2021), PatchTST (Nie et al., 2023), SegRNN (Lin et al., 2025), CycleNet (Lin et al., 2024), STID (Shao et al., 2022), and TimeMixer (Wang et al., 2024a). To ensure a fair comparison, these TSF models are adapted to our UPAP framework, incorporating dual-head outputs and the triple hybrid loss (Eq. 22). For the ELPF baseline, we select Seq2Peak (Zhang et al., 2023), which augments the Transformer backbone by incorporating a parameter-free max-pooling decoder to extract peak-hour series and employs a hybrid loss function to jointly optimize for both global reconstruction and peak intensity. Furthermore, we validate the universal effectiveness of our optimization strategy by explicitly comparing the vanilla versions of these TSF models against their UPAP-enhanced counterparts in the Generality Analysis (Section 4.4).

4.2. Implementation Details

We implement PeakFocus using PyTorch (Paszke et al., 2019) and conduct all experiments on a single NVIDIA A100 GPU. To ensure statistical reliability, all experiments are repeated 5 times with different random seeds, and we report the mean and standard deviation. The model is optimized using the Adam optimizer with an initial learning rate of 10^{-3} . The training process runs for a maximum of 20 epochs with a batch size of 128, employing early stopping with a patience of 5 epochs. As shown in Appendix C and Table 3, the detailed model configurations are provided.

4.3. Main Results Analysis

Table 1 presents the performance comparison. PeakFocus achieves superior performance on WLEL and competitive results on ELC, demonstrating its robustness and adaptability across different forecasting horizons.

Performance on WLEL. PeakFocus demonstrates strong robustness on the WLEL dataset across different horizons. At $H=336$, it achieves a superior $\mathcal{F}_1$. This improves notably compared to the second best baseline Transformer. In contrast, the domain-specific end-to-end baseline Seq2Peak exhibits significantly lower performance. This failure is due to its rigid assumption that peaks are maximum values within fixed 24-hour intervals, causing FPs and missing structural peaks, validating the need for our prominence-based localization strategy. For intensity, PeakFocus reduces the TP-MSE substantially compared to Transformer. At $H=720$, PeakFocus maintains its advantage and improves $\mathcal{F}_1$, further reducing TP-MSE, showing that our model handles intensity smoothing effectively. Global attention models like PatchTST tend to average out peaks. In contrast, PeakFocus uses the LAD module to preserve peak intensity.

Performance on ELC. We observe a trade-off between precision and recall on the ELC dataset. At $H=336$, PeakFocus achieves the best $\mathcal{F}_1$. Similarly, Seq2Peak exhibits suboptimal performance on this benchmark, confirming that max-pooling aggregation cannot adapt to complex peak definitions required for real-world grid safety, whereas PeakFocus bridges this gap effectively. In terms of intensity, STID achieves the lowest TP-MSE but has a low Recall. PeakFocus achieves a much higher Recall, suggesting STID minimizes error by predicting only easy peaks. PeakFocus improves the balanced PIM metric compared to STID. At $H=720$, PeakFocus achieves competitive $\mathcal{F}_1$, ranking second. While SegRNN performs slightly better in this case, PeakFocus outperforms other MLP and Transformer baselines. Regarding intensity, STID achieves lower MSE but misses many peaks. PeakFocus maintains a high Recall, further confirming its effectiveness, prioritizing extreme events over error minimization. This characteristic is vital for ensuring reliability in safety-critical systems.

Table 1. Main results on WLEL and ELC datasets across different horizons (H). We report Recall, Precision, $\mathcal{F}_1$ for peak timing accuracy ($\uparrow$ higher is better), and TP-MSE, TP-MAE for peak intensity error ($\downarrow$ lower is better). BCS and PIM denote balanced metrics. Results are reported as mean $\pm$ std over 5 independent runs. **Bold** indicates the best performance, and underlined indicates the second best.

Dataset	H	Model	Peak Timing			Peak Intensity		Balanced Metrics	
			Recall $\uparrow$	Precision $\uparrow$	$\mathcal{F}_1$ $\uparrow$	TP-MSE $\downarrow$	TP-MAE $\downarrow$	BCS $\downarrow$	PIM $\downarrow$
WLEL	336	CycleNet	0.689 $\pm$ 0.015	<u>0.724$\pm$0.015</u>	0.706 $\pm$ 0.015	0.345 $\pm$ 0.008	0.412 $\pm$ 0.005	0.275 $\pm$ 0.010	1.879 $\pm$ 0.052
		PatchTST	0.628 $\pm$ 0.013	0.681 $\pm$ 0.005	0.653 $\pm$ 0.008	0.420 $\pm$ 0.030	0.467 $\pm$ 0.020	0.321 $\pm$ 0.010	2.143 $\pm$ 0.063
		SegRNN	0.669 $\pm$ 0.003	0.714 $\pm$ 0.003	0.690 $\pm$ 0.003	0.353 $\pm$ 0.015	0.428 $\pm$ 0.011	0.285 $\pm$ 0.005	1.932 $\pm$ 0.025
		STID	0.656 $\pm$ 0.006	0.697 $\pm$ 0.004	0.676 $\pm$ 0.005	0.346 $\pm$ 0.012	0.420 $\pm$ 0.008	0.290 $\pm$ 0.005	1.961 $\pm$ 0.026
		TimeMixer	0.666 $\pm$ 0.022	0.680 $\pm$ 0.007	0.673 $\pm$ 0.014	0.349 $\pm$ 0.016	0.420 $\pm$ 0.008	0.293 $\pm$ 0.010	1.978 $\pm$ 0.058
		Transformer	0.701 $\pm$ 0.010	0.715 $\pm$ 0.010	<u>0.708$\pm$0.010</u>	0.311 $\pm$ 0.016	<u>0.394$\pm$0.007</u>	0.265 $\pm$ 0.006	<u>1.826$\pm$0.030</u>
		Informer	0.633 $\pm$ 0.020	0.659 $\pm$ 0.017	0.646 $\pm$ 0.018	0.509 $\pm$ 0.038	0.516 $\pm$ 0.010	0.345 $\pm$ 0.016	2.303 $\pm$ 0.110
		Seq2Peak	0.307 $\pm$ 0.011	0.516 $\pm$ 0.019	0.385 $\pm$ 0.014	0.418 $\pm$ 0.034	0.456 $\pm$ 0.017	0.455 $\pm$ 0.014	3.599 $\pm$ 0.203
		PeakFocus	0.744$\pm$0.006	0.772$\pm$0.006	0.757$\pm$0.006	0.251$\pm$0.010	0.364$\pm$0.009	0.222$\pm$0.003	1.630$\pm$0.014
	720	CycleNet	0.658 $\pm$ 0.002	0.674 $\pm$ 0.004	0.666 $\pm$ 0.003	0.579 $\pm$ 0.003	0.547 $\pm$ 0.002	0.351 $\pm$ 0.002	2.337 $\pm$ 0.013
		PatchTST	0.603 $\pm$ 0.019	0.644 $\pm$ 0.017	0.622 $\pm$ 0.018	0.775 $\pm$ 0.075	0.651 $\pm$ 0.039	0.407 $\pm$ 0.019	2.811 $\pm$ 0.179
		SegRNN	0.648 $\pm$ 0.026	0.696 $\pm$ 0.015	0.671 $\pm$ 0.021	0.532 $\pm$ 0.037	0.545 $\pm$ 0.021	0.338 $\pm$ 0.018	2.253 $\pm$ 0.120
		STID	0.647 $\pm$ 0.004	0.676 $\pm$ 0.004	0.661 $\pm$ 0.003	0.512 $\pm$ 0.015	0.530 $\pm$ 0.007	0.339 $\pm$ 0.004	2.253 $\pm$ 0.023
		TimeMixer	0.671 $\pm$ 0.035	0.685 $\pm$ 0.012	0.678 $\pm$ 0.023	0.443 $\pm$ 0.018	0.482 $\pm$ 0.009	0.315 $\pm$ 0.015	2.101 $\pm$ 0.092
		Transformer	0.714 $\pm$ 0.015	<u>0.728$\pm$0.015</u>	<u>0.721$\pm$0.014</u>	<u>0.402$\pm$0.036</u>	<u>0.459$\pm$0.021</u>	<u>0.283$\pm$0.009</u>	1.919 $\pm$ 0.048
		Informer	0.629 $\pm$ 0.014	0.663 $\pm$ 0.005	0.646 $\pm$ 0.008	0.702 $\pm$ 0.070	0.630 $\pm$ 0.029	0.383 $\pm$ 0.010	2.596 $\pm$ 0.090
		Seq2Peak	0.256 $\pm$ 0.001	0.437 $\pm$ 0.003	0.323 $\pm$ 0.012	0.438 $\pm$ 0.125	0.504 $\pm$ 0.033	0.491 $\pm$ 0.021	4.321 $\pm$ 0.072
		PeakFocus	0.754$\pm$0.006	0.776$\pm$0.005	0.764$\pm$0.005	0.318$\pm$0.027	0.413$\pm$0.014	0.238$\pm$0.010	1.702$\pm$0.045
ELC	336	CycleNet	0.419 $\pm$ 0.018	0.418 $\pm$ 0.021	0.418 $\pm$ 0.019	0.699 $\pm$ 0.015	0.653 $\pm$ 0.006	0.497 $\pm$ 0.008	3.974 $\pm$ 0.154
		PatchTST	0.684 $\pm$ 0.014	<u>0.736$\pm$0.018</u>	0.709 $\pm$ 0.016	1.037 $\pm$ 0.039	0.778 $\pm$ 0.014	0.400 $\pm$ 0.008	2.835 $\pm$ 0.068
		SegRNN	0.606 $\pm$ 0.026	0.651 $\pm$ 0.028	0.628 $\pm$ 0.027	0.811 $\pm$ 0.031	0.691 $\pm$ 0.012	0.410 $\pm$ 0.011	2.845 $\pm$ 0.096
		STID	0.366 $\pm$ 0.032	0.387 $\pm$ 0.035	0.377 $\pm$ 0.034	0.579$\pm$0.028	0.602$\pm$0.010	0.495 $\pm$ 0.012	4.112 $\pm$ 0.299
		TimeMixer	<u>0.693$\pm$0.007</u>	0.729 $\pm$ 0.011	<u>0.711$\pm$0.008</u>	0.850 $\pm$ 0.060	0.705 $\pm$ 0.027	<u>0.374$\pm$0.011</u>	<u>2.568$\pm$0.094</u>
		Transformer	0.382 $\pm$ 0.111	0.314 $\pm$ 0.095	0.345 $\pm$ 0.102	0.726 $\pm$ 0.114	0.680 $\pm$ 0.046	0.537 $\pm$ 0.034	5.266 $\pm$ 1.481
		Informer	0.541 $\pm$ 0.017	0.423 $\pm$ 0.018	0.475 $\pm$ 0.018	0.703 $\pm$ 0.012	0.639 $\pm$ 0.004	0.469 $\pm$ 0.010	3.518 $\pm$ 0.135
		Seq2Peak	0.349 $\pm$ 0.015	0.503 $\pm$ 0.022	0.412 $\pm$ 0.018	0.668 $\pm$ 0.056	0.622 $\pm$ 0.026	0.494 $\pm$ 0.003	3.954 $\pm$ 0.053
		PeakFocus	0.723$\pm$0.004	0.780$\pm$0.004	0.751$\pm$0.004	0.930 $\pm$ 0.100	0.735 $\pm$ 0.045	0.365$\pm$0.012	2.537$\pm$0.123
	720	CycleNet	0.435 $\pm$ 0.009	0.427 $\pm$ 0.010	0.431 $\pm$ 0.009	<u>0.726$\pm$0.008</u>	0.668 $\pm$ 0.003	0.495 $\pm$ 0.003	3.915 $\pm$ 0.064
		PatchTST	0.674 $\pm$ 0.026	0.692 $\pm$ 0.057	0.683 $\pm$ 0.041	1.004 $\pm$ 0.059	0.782 $\pm$ 0.026	0.409 $\pm$ 0.018	2.902 $\pm$ 0.144
		SegRNN	0.725$\pm$0.012	0.778$\pm$0.013	0.750$\pm$0.012	0.793 $\pm$ 0.043	0.686 $\pm$ 0.019	0.346$\pm$0.008	2.358$\pm$0.059
		STID	0.457 $\pm$ 0.028	0.480 $\pm$ 0.032	0.468 $\pm$ 0.030	0.652$\pm$0.009	0.639 $\pm$ 0.003	0.463 $\pm$ 0.014	3.467 $\pm$ 0.217
		TimeMixer	0.674 $\pm$ 0.033	0.678 $\pm$ 0.054	0.676 $\pm$ 0.043	0.840 $\pm$ 0.084	0.713 $\pm$ 0.040	0.390 $\pm$ 0.022	<u>2.694$\pm$0.196</u>
		Transformer	0.530 $\pm$ 0.107	0.420 $\pm$ 0.079	0.469 $\pm$ 0.090	0.929 $\pm$ 0.128	0.755 $\pm$ 0.050	0.505 $\pm$ 0.042	4.194 $\pm$ 0.874
		Informer	0.334 $\pm$ 0.171	0.244 $\pm$ 0.125	0.282 $\pm$ 0.145	0.727 $\pm$ 0.371	0.607$\pm$0.305	0.549 $\pm$ 0.027	4.237 $\pm$ 0.788
		Seq2Peak	0.370 $\pm$ 0.008	0.535 $\pm$ 0.011	0.437 $\pm$ 0.009	0.785 $\pm$ 0.119	0.669 $\pm$ 0.049	0.500 $\pm$ 0.016	3.988 $\pm$ 0.217
		PeakFocus	<u>0.711$\pm$0.009</u>	<u>0.754$\pm$0.014</u>	<u>0.732$\pm$0.011</u>	1.169 $\pm$ 0.077	0.842 $\pm$ 0.040	0.403 $\pm$ 0.010	2.925 $\pm$ 0.118

4.4. Ablation Study

We perform component analysis to validate our proposed architecture and optimization strategy. Table 2 examines the impact of MSM-PL and LAD, while Table 4 verifies the universality of the UPAP framework across diverse baselines.

Observation 1: Multi Scale Mixing Filters Local Fluctuations. The w/o MSM-PL variant employs a simple linear layer instead of the hierarchical structure. As shown in Table 2, on the WLEL dataset, the $\mathcal{F}_1$ score drops by approximately 13%, accompanied by a significant decline in Precision. This indicates that the linear layer fails to ef-

fectively filter out local fluctuations. In contrast, MSM-PL successfully distinguishes true peaks from background disturbances, ensuring robust localization.

Observation 2: Explicit Timing Context Improves Intensity. W/o LAD variant removes the peak hidden state injection, relying on the general context for regression. While the localization performance remains relatively stable, the intensity error TP-MSE on WLEL increases by roughly 34% (Table 2). This result confirms the critical importance of location awareness. Without explicit timing guidance from the peak branch, the decoder lacks focus and tends to underestimate peak intensity due to global smoothing.

Table 2. Ablation results on WLEL and ELC datasets. w/o MSM-PL replaces MSM-PL with a linear layer; w/o LAD uses $\mathbf{X}_{\text{en,out}}$ as Keys/Values instead of Context Gate Fusion outputs, lacking peak timing context; w/o UPAP removes the triple hybrid loss, training with standard MSE only. Data is presented as mean $\pm$ std. **Bold** indicates the best performance, and underlined indicates the second best.

Dataset	H	Variant	Peak Timing			Peak Intensity		Balanced Metrics	
			Recall $\uparrow$	Precision $\uparrow$	$\mathcal{F}_1$ $\uparrow$	TP-MSE $\downarrow$	TP-MAE $\downarrow$	BCS $\downarrow$	PIM $\downarrow$
WLEL	336	PeakFocus	0.744 $\pm$ 0.006	0.772$\pm$0.006	0.757$\pm$0.006	0.251$\pm$0.010	0.364$\pm$0.009	0.222$\pm$0.003	1.630$\pm$0.014
		w/o MSM-PL	0.633 $\pm$ 0.033	0.689 $\pm$ 0.003	0.659 $\pm$ 0.017	0.362 $\pm$ 0.035	0.443 $\pm$ 0.020	0.303 $\pm$ 0.012	2.036 $\pm$ 0.069
		w/o LAD	0.744$\pm$0.002	<u>0.768$\pm$0.004</u>	<u>0.756$\pm$0.003</u>	<u>0.336$\pm$0.011</u>	<u>0.425$\pm$0.011</u>	<u>0.248$\pm$0.003</u>	<u>1.743$\pm$0.016</u>
		w/o UPAP	0.524 $\pm$ 0.029	0.591 $\pm$ 0.021	0.552 $\pm$ 0.026	0.409 $\pm$ 0.020	0.470 $\pm$ 0.013	0.341 $\pm$ 0.012	2.893 $\pm$ 0.143
	720	PeakFocus	0.754$\pm$0.006	0.776$\pm$0.005	0.764$\pm$0.005	0.318$\pm$0.027	0.413$\pm$0.014	0.238$\pm$0.010	1.702$\pm$0.045
		w/o MSM-PL	0.668 $\pm$ 0.014	0.693 $\pm$ 0.021	0.680 $\pm$ 0.013	0.491 $\pm$ 0.058	0.524 $\pm$ 0.035	0.324 $\pm$ 0.016	2.163 $\pm$ 0.108
		w/o LAD	<u>0.752$\pm$0.006</u>	<u>0.772$\pm$0.005</u>	<u>0.762$\pm$0.005</u>	<u>0.469$\pm$0.061</u>	<u>0.497$\pm$0.022</u>	<u>0.278$\pm$0.014</u>	<u>1.903$\pm$0.078</u>
		w/o UPAP	0.506 $\pm$ 0.025	0.547 $\pm$ 0.036	0.523 $\pm$ 0.030	0.532 $\pm$ 0.047	0.539 $\pm$ 0.029	0.380 $\pm$ 0.022	3.240 $\pm$ 0.323
ELC	336	PeakFocus	0.723$\pm$0.004	0.780$\pm$0.004	0.751$\pm$0.004	0.930 $\pm$ 0.100	0.735 $\pm$ 0.045	0.365$\pm$0.012	2.537$\pm$0.123
		w/o MSM-PL	0.572 $\pm$ 0.009	0.410 $\pm$ 0.022	0.477 $\pm$ 0.016	0.701$\pm$0.045	0.654$\pm$0.026	0.467 $\pm$ 0.007	3.491 $\pm$ 0.095
		w/o LAD	<u>0.621$\pm$0.061</u>	<u>0.619$\pm$0.095</u>	<u>0.619$\pm$0.079</u>	<u>0.742$\pm$0.075</u>	<u>0.660$\pm$0.040</u>	<u>0.403$\pm$0.037</u>	<u>2.813$\pm$0.375</u>
		w/o UPAP	0.436 $\pm$ 0.053	0.462 $\pm$ 0.059	0.448 $\pm$ 0.056	0.818 $\pm$ 0.047	0.696 $\pm$ 0.023	0.460 $\pm$ 0.030	5.455 $\pm$ 1.328
	720	PeakFocus	0.711$\pm$0.009	0.754$\pm$0.014	0.732$\pm$0.011	1.169 $\pm$ 0.077	0.842 $\pm$ 0.040	0.403$\pm$0.010	2.925 $\pm$ 0.118
		w/o MSM-PL	0.588 $\pm$ 0.014	0.384 $\pm$ 0.018	0.464 $\pm$ 0.014	0.754$\pm$0.113	0.682$\pm$0.059	0.482 $\pm$ 0.018	3.702 $\pm$ 0.236
		w/o LAD	<u>0.683$\pm$0.012</u>	<u>0.648$\pm$0.032</u>	<u>0.665$\pm$0.022</u>	<u>0.903$\pm$0.046</u>	<u>0.746$\pm$0.025</u>	<u>0.405$\pm$0.013</u>	2.824$\pm$0.116
		w/o UPAP	0.486 $\pm$ 0.013	0.514 $\pm$ 0.015	0.499 $\pm$ 0.014	1.063 $\pm$ 0.066	0.822 $\pm$ 0.027	0.476 $\pm$ 0.012	4.611 $\pm$ 0.413

Observation 3: Unified Optimization Bridges Localization and Regression. We first evaluate the necessity of the UPAP strategy within our proposed framework. As presented in Table 2, removing the loss of triple hybrid (w/ no UPAP) and training solely with standard MSE leads to a severe performance degradation. Specifically, the $\mathcal{F}_1$ score drops significantly, and the intensity error increases by a large margin. This deterioration occurs because the standard global reconstruction objective is dominated by the majority of non-peak data, causing the model to neglect sparse extreme events. In contrast, UPAP effectively counteracts this by enforcing synchronized supervision on both timing and intensity. Furthermore, this optimization strategy is model-agnostic; as detailed in Appendix D, applying UPAP to other TSF baselines consistently enhances their peak forecasting capability, verifying its universality.

4.5. Qualitative Visualization

We conduct a qualitative comparison between PeakFocus and PatchTST to visually verify our proposed modules. Figure 3 illustrates the forecasting trajectories on the WLEL dataset during two distinct periods characterized by high-frequency volatility and extreme load events.

Analysis of Localization Precision (Figures 3a vs. 3b). We first examine the capability of peak timing localization using the sample from April 12 to April 26, 2025. As shown in Figure 3a, PatchTST exhibits severe limitations in distinguishing structural peaks from random noise. It generates numerous False Positives (indicated by red crosses) in non-

peak regions and suffers from temporal misalignment. This failure stems from its reliance on global point-wise reconstruction, which lacks specific sensitivity to local prominence. In sharp contrast, Figure 3b demonstrates that PeakFocus achieves exceptional precision. The model accurately locks onto the true peak moments, significantly reducing false alarms. This improvement serves as direct evidence for the effectiveness of our MSM-PL. By leveraging the hierarchical pyramid structure to filter out high-frequency disturbances and the cascade injection mechanism to refine temporal resolution, PeakFocus successfully resolves the trade-off between misjudgment and misalignment, ensuring reliable scheduling guidance for grid operators.

Analysis of Intensity Estimation (Figures 3c vs. 3d). Next, we evaluate the performance of peak magnitude regression using the sample from June 21 to July 05, 2025. Figure 3c highlights the intensity smoothing phenomenon inherent in standard TSF models. PatchTST produces a flattened trajectory that systematically underestimates the magnitude of extreme events. This is because its global objective function treats sparse peaks as outliers, diluting them towards the mean trend. Conversely, Figure 3d shows that PeakFocus recovers the full amplitude of these extreme surges, aligning closely with the ground truth intensity. This capability is attributed to the LAD. By explicitly conditioning the regression process on the peak timing context extracted by the locator, the LAD provides the necessary guidance to counteract smoothing biases, allowing the model to aggressively predict high-magnitude values exactly when they occur.

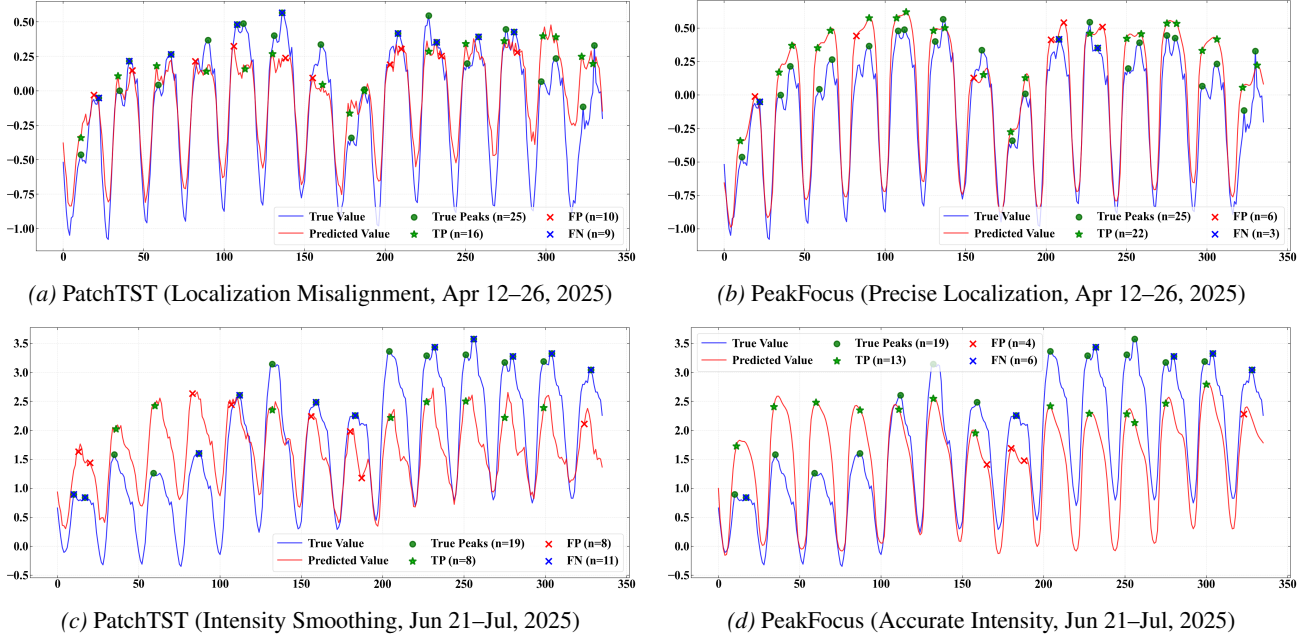


Figure 3. Qualitative comparison on the WLEL dataset. (a) and (b) compare peak localization performance, showing PatchTST’s temporal misalignment versus PeakFocus’s high precision. (c) and (d) compare intensity estimation, highlighting PatchTST’s severe underestimation bias versus PeakFocus’s robust ability to capture extreme load surges.

5. Related Works

5.1. Time Series Forecasting

TSF plays a pivotal role in various domains, evolving from Recurrent Neural Networks (RNNs) to MLPs and Transformers (Yu et al., 2025; Zhu et al., 2025a; Cheng et al., 2025). RNN-based approaches, SegRNN (Lin et al., 2025) utilizes recurrent units for efficient long-term modeling, while P-sLSTM (Kong et al., 2025) improves upon LSTM by incorporating patching and channel independence. Meanwhile, MLP-based models, including DLinear (Zeng et al., 2023), STID (Shao et al., 2022), and TimeMixer (Wang et al., 2024a), achieve strong performance via simple spatial-temporal decomposition. Transformers like Informer (Zhou et al., 2021) and Autoformer (Wu et al., 2021) reduce complexity using sparse attention, whereas PatchTST (Nie et al., 2023) and iTransformer (Liu et al., 2024) improve results through patch-based tokenization and inverted attention. However, these general TSF models are often suboptimal for the rigorous demands of ELPF. Since peaks constitute sparse tail events in the load distribution, standard objectives drive models to learn a conservative representation biased towards the mean to minimize aggregate penalty. Consequently, critical sharp spikes are effectively treated as noise and smoothed out, sacrificing the detection of safety-critical extremes. (Hu et al., 2025b). Even multi-scale models like TimeMixer (Wang et al., 2024a) and AMD (Hu et al., 2025a) prioritize global trend decomposition, lacking the fine-grained resolution required for precise peak timing.

5.2. Electricity Load Peak Forecasting

Methodologies have shifted from heuristic two-stage pipelines to unified end-to-end frameworks (Amara-Ouali et al., 2023; Dai et al., 2021). Conventional methods treat detection as post-processing, optimizing features for average error rather than peak discrimination. End-to-end approaches bridge this gap. Seq2Peak (Zhang et al., 2023) uses a max-pooling decoder, while Seq2LPP (Zhu et al., 2025b) adds weather-guided attention. However, defining peaks as fixed-interval maxima reduces ELPF to downsampling, failing to capture structural peaks from local topology. Furthermore, they lack multi-scale mechanisms to reconcile fine-grained fluctuations with coarse-grained shifts. Second, intensity regression is insufficiently conditioned on peak timing. Without this context, regression suffers from smoothing bias and intensity underestimation.

6. Conclusion

PeakFocus bridges temporal localization and intensity regression in ELPF via a unified end-to-end framework. MSM-PL resolves misjudgment via hierarchical fusion, while LAD prevents intensity dilution using timing contexts. Extensive experiments on the ELC dataset and the WLEL dataset confirm PeakFocus establishes a new SOTA. A limitation is the reliance on historical sequences, ignoring external covariates and multivariate dependencies. Future work will address this by extending the framework to multivariate forecasting and integrating external covariates.

Impact Statement

This paper presents PeakFocus to bridge the gap between TSF and ELPF. Unlike standard methods that focus on overall reconstruction, our work targets the peak events that determine grid stability. This offers a practical solution for the operational needs of real power systems. The impact of this work covers grid reliability and environmental sustainability. By improving the accuracy of peak timing and intensity predictions, PeakFocus helps operators optimize resource dispatch and Demand Response. This reduces reliance on carbon-intensive peaker plants and supports the integration of renewable energy sources. Consequently, our method lowers the carbon footprint of energy systems and aids global carbon neutrality goals. However, power grids are safety-critical infrastructure. Deep learning models may struggle with data shifts caused by extreme climate events. Relying on automated forecasting risks false negatives which could lead to power shortages. Therefore, PeakFocus serves as a decision-support tool for human experts. It requires human verification and should not act as a fully autonomous control system.

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A. Peak Labeling Algorithm Details

To generate ground-truth peak labels $\mathbf{Y}_{t,\text{true}}$, we employ a peak detection algorithm adapted from the `findpeaks` library. This method ensures identification robustness by tracking local extrema governed by two key hyperparameters: a sensitivity threshold η and a lookahead window ℓ . The algorithm iterates through the sequence maintaining a dynamic maximum (mx). A candidate peak is confirmed only when the intensity drops by at least η from the current mx . The lookahead parameter ℓ reinforces local optimality by ensuring the detected peak dominates its immediate temporal neighborhood.

Algorithm 1 Heuristic Peak Detection (`findpeaks` Implementation)

```

1: Input: Load sequence  $\mathbf{Y} \in \mathbb{R}^H$ , Sensitivity threshold  $\eta$ , Lookahead parameter  $\ell$ .
2: Output: Set of peak indices  $\mathcal{P}$ .
3:  $\mathcal{P} \leftarrow \emptyset$ 
4:  $mx \leftarrow -\infty, mn \leftarrow +\infty$ 
5:  $pos_{mx} \leftarrow 0$ 
6:  $search\_state \leftarrow \text{FindMax}$ 
7: for  $t = 1$  to  $H$  do
8:    $v \leftarrow \mathbf{Y}[t]$ 
9:   {Update dynamic extrema}
10:  if  $v > mx$  then
11:     $mx \leftarrow v; pos_{mx} \leftarrow t$ 
12:  end if
13:  if  $v < mn$  then
14:     $mn \leftarrow v$ 
15:  end if
16:  if  $search\_state == \text{FindMax}$  then
17:    if  $v < mx - \eta$  then
18:      {Peak Candidate Found: Significant drop observed}
19:      {Explicit Lookahead Check: Ensure strictly local optimality}
20:       $is\_valid \leftarrow \text{True}$ 
21:      for  $k = 1$  to  $\ell$  do
22:        if  $pos_{mx} + k \leq H$  and  $\mathbf{Y}[pos_{mx} + k] > mx$  then
23:           $is\_valid \leftarrow \text{False}$ 
24:        break
25:      end if
26:    end for
27:    if  $is\_valid$  then
28:      Add  $pos_{mx}$  to  $\mathcal{P}$ 
29:       $mn \leftarrow v$  {Reset min to current value}
30:       $search\_state \leftarrow \text{FindMin}$ 
31:    else
32:      {Lookahead failed: Update max to the superior neighbor}
33:       $mx \leftarrow \mathbf{Y}[pos_{mx} + k]$ 
34:       $pos_{mx} \leftarrow pos_{mx} + k$ 
35:    end if
36:  end if
37: else
38:   if  $v > mn + \eta$  then
39:     {Valley Confirmed: Significant rise observed}
40:      $mx \leftarrow v$  {Reset max to current value}
41:      $pos_{mx} \leftarrow t$ 
42:      $search\_state \leftarrow \text{FindMax}$ 
43:   end if
44: end if
45: end for
46: return  $\mathcal{P}$ 

```

B. Tolerance-based Condense-and-Match Protocol

As outlined in Section 3.4, evaluating peak forecasting requires a rigorous mapping from continuous probability outputs to discrete event pairs. Since the model output $\mathbf{Y}_{t\text{-pred}}$ typically exhibits contiguous high-probability regions rather than single spikes, a direct thresholding approach would yield redundant positives. We therefore employ a two-phase protocol:

(1) Condensation Phase: We first binarize the probability sequence using a confidence threshold τ . Contiguous temporal segments of 1s are treated as single event clusters. For each cluster, we extract a single discrete representative $\hat{t}$ by selecting the time step with the maximum probability, forming the candidate set $\hat{\mathcal{P}}$. This ensures that wide probability peaks are condensed into precise temporal locations.

(2) Matching Phase: We perform a bipartite matching between the candidate set $\hat{\mathcal{P}}$ and the ground-truth set $\mathcal{P}$. A candidate $\hat{t}$ is considered a valid match for a ground truth t if it falls within the tolerance window (i.e., $|\hat{t} - t| \leq \delta$). To resolve conflicts where multiple candidates compete for a single ground truth (or vice versa), we employ a greedy strategy that prioritizes pairs with the smallest temporal distance $|t - \hat{t}|$. This enforces a strict one-to-one pairing, preventing a single prediction from being credited for multiple ground-truth peaks.

Algorithm 2 Condense-and-Match Under Tolerance

```

1: Input: Probabilities  $\mathbf{Y}_{t\text{-pred}} \in [0, 1]^H$ , Threshold  $\tau$ , Tolerance  $\delta$ .
2: Output: Metrics (TP, FP, FN) and Match Set  $\mathcal{M}$ .
3: {Phase 1: Condense Probabilities to Indices}
4:  $\mathbf{b} \leftarrow \mathbb{I}(\mathbf{Y}_{t\text{-pred}} \geq \tau)$  {Binarize predictions}
5:  $\hat{\mathcal{P}} \leftarrow \emptyset$ 
6:  $t \leftarrow 1$ 
7: while  $t \leq H$  do
8:   if  $b_t = 0$  then
9:      $t \leftarrow t + 1$ 
10:  else
11:     $t_s \leftarrow t$  {Start of high-probability region}
12:    while  $t \leq H$  and  $b_t = 1$  do
13:       $t \leftarrow t + 1$ 
14:    end while
15:     $t_e \leftarrow t - 1$ 
16:     $\hat{t} \leftarrow \arg \max_{u \in [t_s, t_e]} (\mathbf{Y}_{t\text{-pred}})_u$  {Select max probability as representative}
17:    Append  $\hat{t}$  to  $\hat{\mathcal{P}}$ 
18:  end if
19: end while
20: {Phase 2: Bipartite Matching}
21:  $\mathcal{P} \leftarrow \{t \mid \mathbf{Y}_{t\text{-true}}[t] = 1\}$  {Ground-truth indices}
22:  $\mathcal{C} \leftarrow \emptyset$  {Candidate pairs}
23: for all  $t \in \mathcal{P}$  do
24:   for all  $\hat{t} \in \hat{\mathcal{P}}$  do
25:     if  $|t - \hat{t}| \leq \delta$  then
26:       Add  $(t, \hat{t}, |t - \hat{t}|)$  to  $\mathcal{C}$ 
27:     end if
28:   end for
29: end for
30: Sort  $\mathcal{C}$  by distance  $d$  in ascending order {Greedy strategy}
31:  $\mathcal{M} \leftarrow \emptyset$ 
32: Mark all  $t \in \mathcal{P}$  and  $\hat{t} \in \hat{\mathcal{P}}$  as unmatched
33: for all  $(t, \hat{t}, d) \in \mathcal{C}$  do
34:   if  $t$  is unmatched and  $\hat{t}$  is unmatched then
35:     Add  $(t, \hat{t})$  to  $\mathcal{M}$ 
36:     Mark  $t$  and  $\hat{t}$  as matched
37:   end if
38: end for
39: TP  $\leftarrow |\mathcal{M}|$ 
40: FP  $\leftarrow |\hat{\mathcal{P}}| - \text{TP}$ 
41: FN  $\leftarrow |\mathcal{P}| - \text{TP}$ 
42: return TP, FP, FN,  $\mathcal{M}$ 

```

C. Detailed Implementation Settings

Training Dynamics & Loss Function. The loss scaling factors in the triple hybrid objective function (Eq. 22) are empirically set as follows: $\lambda_1 = 0.4$ for the Global Loss, $\lambda_2 = 0.2$ for the TP-MSE Loss, and $\lambda_3 = 0.4$ for the Localization Loss. Other optimization settings, including the learning rate scheduler and batch size, are detailed in the left section of Table 3.

Model Architecture. We set the dimension of the latent representation d_{model} to 256. The encoder utilizes a Residual MLP backbone with 2 layers. For the MSM-PL (Eq. 7), we employ a two-stage downsampling pyramid ($K = 2$). The pooling operations at levels $k = 1, 2$ are configured with specific kernel sizes $\kappa^{(k)}$ and strides $\sigma^{(k)}$ as listed in Table 3. The feature extractors ψ_k utilize 1D convolutions with a kernel size $\kappa_c = 3$ and GELU activation. The LAD employs a feed-forward dimension d_{ff} of 1024 ($4 \times d_{\text{model}}$) and utilizes standard ReLU activation.

Table 3. Detailed Hyperparameter Settings for PeakFocus.

Training Dynamics & Loss Function		Model Architecture & Inference	
Input Sequence Length (L)	168	Model Dimension (d_{model})	256
Prediction Horizon (H)	{336, 720}	Feed-Forward Dim (d_{ff})	1024
Batch Size	128	Attention Heads (n_{heads})	4
Learning Rate	0.001	Encoder Layers (e_{layers})	1
Optimizer	Adam	MLP Layers (Backbone)	2
Training Epochs	20	Dropout	0.1
Early Stopping Patience	5	Activation (LAD / MSM-PL)	ReLU / GELU
LR Scheduler	CosineAnnealingLR	MSM-PL Pooling $k = 1$ ($\kappa^{(1)} / \sigma^{(1)}$)	3 / 2
Model Saving Criterion	Lowest BCS	MSM-PL Pooling $k = 2$ ($\kappa^{(2)} / \sigma^{(2)}$)	5 / 2
Global Loss Weight (λ_1)	0.4	MSM-PL Conv Kernel κ_c	3
TP-MSE Loss Weight (λ_2)	0.2	Peak Lookahead (ℓ)	5 (ELC) / 3 (WLEL)
Localization Loss Weight (λ_3)	0.4	Peak Tolerance (δ)	1
Gaussian Kernel Width (γ)	1.0	Sensitivity Threshold (η)	0.5
Probability Threshold (τ)	0.5	PIM Constant (ϵ)	0.01
BCS Weight (α)	0.5		

D. Generality Analysis Experiments

This section provides a detailed empirical validation of the universality of the UPAP, complementing Observation 3 in the main text. To verify that the benefits of UPAP extend beyond the PeakFocus architecture, we integrate the proposed dual-head framework and Triple Hybrid Loss into four representative state-of-the-art TSF baselines: CycleNet, STID, TimeMixer, and Transformer.

We compare the Original vanilla models (trained with standard global MSE loss) against their + UPAP counterparts. Notably, for the Transformer backbone, we introduce an additional comparison with Seq2Peak. Since Seq2Peak is fundamentally a Transformer-based model optimized with a max-pooling strategy, placing it alongside the Transformer variants offers a direct evaluation of different optimization strategies on the same backbone. As shown in Table 4, the results on the WLEL dataset reveal key insights:

- Seq2Peak exhibits significantly inferior performance compared to even the Original vanilla Transformer. Its rigid max-pooling mechanism inherently assumes a simplified peak distribution, rendering it unable to adapt to the complex multi-peak patterns frequently observed in real-world scenarios.
- In contrast, the + UPAP variants consistently outperform both the Original baselines and the domain-specific Seq2Peak. Specifically, applying UPAP to the Transformer yields the highest Recall and lowest TP-MSE. This demonstrates that UPAP serves as an effective plug-and-play optimization strategy, empowering general architectures to capture extreme events better than specialized but rigid alternatives.

Table 4. Generality analysis of the proposed UPAP. We apply the UPAP framework to various baselines. For the **Transformer**, we explicitly compare three variants: the Original vanilla version, the domain-specific Seq2Peak model (which shares the Transformer backbone), and our + UPAP version. Data is presented as mean $\pm$ std. **Bold** indicates the best performance within each model group.

H	Model	Variant	Peak Timing			Peak Intensity		Balanced Metrics	
			Recall $\uparrow$	Precision $\uparrow$	$\mathcal{F}_1 \uparrow$	TP-MSE $\downarrow$	TP-MAE $\downarrow$	BCS $\downarrow$	PIM $\downarrow$
336	CycleNet	Original	0.623 $\pm$ 0.008	0.666 $\pm$ 0.007	0.640 $\pm$ 0.006	0.406 $\pm$ 0.003	0.455 $\pm$ 0.002	0.291 $\pm$ 0.003	2.346 $\pm$ 0.028
		+ UPAP	0.689$\pm$0.015	0.724$\pm$0.015	0.706$\pm$0.015	0.345$\pm$0.008	0.412$\pm$0.005	0.275$\pm$0.010	1.879$\pm$0.052
	STID	Original	0.652 $\pm$ 0.005	0.677 $\pm$ 0.006	0.660 $\pm$ 0.005	0.439 $\pm$ 0.013	0.485 $\pm$ 0.010	0.292 $\pm$ 0.002	2.392 $\pm$ 0.026
		+ UPAP	0.656$\pm$0.006	0.697$\pm$0.004	0.676$\pm$0.005	0.346$\pm$0.012	0.420$\pm$0.008	0.290$\pm$0.005	1.961$\pm$0.026
	TimeMixer	Original	0.544 $\pm$ 0.015	0.558 $\pm$ 0.015	0.548 $\pm$ 0.015	0.395 $\pm$ 0.018	0.458 $\pm$ 0.006	0.337 $\pm$ 0.007	2.772 $\pm$ 0.107
		+ UPAP	0.666$\pm$0.022	0.680$\pm$0.007	0.673$\pm$0.014	0.349$\pm$0.016	0.420$\pm$0.008	0.293$\pm$0.010	1.978$\pm$0.058
	Transformer	Original	0.670 $\pm$ 0.012	0.718$\pm$0.008	0.689 $\pm$ 0.010	0.316 $\pm$ 0.020	0.412 $\pm$ 0.012	0.253$\pm$0.006	2.005 $\pm$ 0.045
		Seq2Peak	0.307 $\pm$ 0.011	0.516 $\pm$ 0.019	0.385 $\pm$ 0.014	0.418 $\pm$ 0.034	0.456 $\pm$ 0.017	0.455 $\pm$ 0.014	3.599 $\pm$ 0.203
		+ UPAP	0.701$\pm$0.010	0.715 $\pm$ 0.010	0.708$\pm$0.010	0.311$\pm$0.016	0.394$\pm$0.007	0.265 $\pm$ 0.006	1.826$\pm$0.030
	720	CycleNet	Original	0.616 $\pm$ 0.011	0.652 $\pm$ 0.005	0.630 $\pm$ 0.007	0.668 $\pm$ 0.005	0.590 $\pm$ 0.004	0.337$\pm$0.003
+ UPAP			0.658$\pm$0.002	0.674$\pm$0.004	0.666$\pm$0.003	0.579$\pm$0.003	0.547$\pm$0.002	0.351 $\pm$ 0.002	2.337$\pm$0.013
STID		Original	0.616 $\pm$ 0.005	0.621 $\pm$ 0.006	0.615 $\pm$ 0.005	0.682 $\pm$ 0.043	0.622 $\pm$ 0.025	0.354 $\pm$ 0.007	3.088 $\pm$ 0.146
		+ UPAP	0.647$\pm$0.004	0.676$\pm$0.004	0.661$\pm$0.003	0.512$\pm$0.015	0.530$\pm$0.007	0.339$\pm$0.004	2.253$\pm$0.023
TimeMixer		Original	0.513 $\pm$ 0.007	0.512 $\pm$ 0.010	0.511 $\pm$ 0.008	0.459 $\pm$ 0.021	0.499 $\pm$ 0.008	0.372 $\pm$ 0.003	3.021 $\pm$ 0.029
		+ UPAP	0.671$\pm$0.035	0.685$\pm$0.012	0.678$\pm$0.023	0.443$\pm$0.018	0.482$\pm$0.009	0.315$\pm$0.015	2.101$\pm$0.092
Transformer		Original	0.655 $\pm$ 0.012	0.701 $\pm$ 0.016	0.673 $\pm$ 0.014	0.445 $\pm$ 0.035	0.492 $\pm$ 0.016	0.287 $\pm$ 0.010	2.203 $\pm$ 0.101
		Seq2Peak	0.256 $\pm$ 0.001	0.437 $\pm$ 0.003	0.323 $\pm$ 0.012	0.438 $\pm$ 0.125	0.504 $\pm$ 0.033	0.491 $\pm$ 0.021	4.321 $\pm$ 0.072
		+ UPAP	0.714$\pm$0.015	0.728$\pm$0.015	0.721$\pm$0.014	0.402$\pm$0.036	0.459$\pm$0.021	0.283$\pm$0.009	1.919$\pm$0.048